Weikai Lin

CONTACT INFORMATION

Ph.D. Student
University of Rochester
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Rochester, NY 14627

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https://linwk20.github.io/
Google Scholar | LinkedIn

RESEARCH INTERESTS

I work at the intersection of computer graphics, computer architecture, and imaging systems. My research bridges the gap between hardware and software through system-level co-optimization. Current themes:

- **AR/VR neural rendering and acceleration:** neural rendering algorithms for AR/VR and their hardware accelerators. [CVPR'26, SIGGRAPH Asia'25, ASPLOS'25a, ASPLOS'25b, ISCA'25, HotMobile'25, TACO'24]
- **Imaging system design:** co-design of sensors and optics with downstream vision models. [DAC'26, DAC'25, WACV'25]

EDUCATION

University of Rochester, Rochester, NY, USA
Ph.D., Computer Science — Aug 2023 – May 2028 (expected)
M.Sc., Computer Science — Aug 2023 – May 2025
Advisor: Yuhao Zhu

Peking University, Beijing, China
M.Sc., Intelligence Science and Technology — Sep 2020 – Jun 2023
Advisor: Dingsheng Luo

Tsinghua University, Beijing, China
B.E., Electronic Engineering — Aug 2016 – Jun 2020
Advisor: Li Su

HONORS AND AWARDS

- **2026, ACM/IEEE DAC Young Fellow**
- **2026, ACM ASPLOS Travel Grant**
- **2025, ACM ASPLOS Best Paper Award**
  "MetaSapiens: Real-Time Neural Rendering with Efficiency-Aware Pruning and Accelerated Foveated Rendering" (first author)
- **2025, OpenAI Research Grant ($5K)**
- **2025, ACM ASPLOS Travel Grant**
- **2022, Baosteel's Outstanding Student Award for HMT Students**
- **2022, First Prize, Peking University Challenge Cup**
- **2021, First Class Scholarship for Taiwan Master's Students**

PROFESSIONAL POSITIONS

Meta, California, USA
Research Scientist Intern — May 2026 – Oct 2026

Pixocial AI Lab, Bellevue, WA, USA
Applied Research Intern — May 2025 – Aug 2025
Mentor: Haoxiang Li.
Topics: Physics-based video diffusion and dynamic hair rendering. [project page]

University of Rochester, Rochester, NY, USA
Graduate Researcher — Aug 2023 – present
Teaching Assistant — Fall 2024, Spring 2025, Fall 2025

Advanced Micro Devices, Inc. (AMD), Beijing, China
Research Intern — Jul 2022 – Feb 2023
Mentor: Fuwei Yang.
Topics: SLAM and DNN evaluations on AMD GPUs. Participated in a ROCm release.

Peking University, Beijing, China
Graduate Researcher — Sep 2020 – Jun 2023
Teaching Assistant — 2021 – 2022
Topics: Robotics, robot learning, and autonomous driving.

Chinese Academy of Sciences (CAS), Beijing, China
Engineer Intern
Topics: HLS-based FPGA wireless communication system development. — 2021 – 2022

Tsinghua University, Beijing, China
Undergraduate Researcher — Sep 2019 – Jun 2020
Mentor: Li Su.
Topics: FPGA-based wireless codec system development.

Cambricon Technologies Co., Ltd., Beijing, China
Research Intern — Jul 2019 – Aug 2019
Mentor: Miao Li.
Topics: Neural network pruning.

PUBLICATIONS

* denotes equal contribution.

- Ruofan Xing, Arman Akbari, Weikai Lin, Adith Boloor, Alexander Montes McNeil, Michael Moebius, Yongmin Liu, Yuhao Zhu, Xuan Zhang
  "HoloCode: Hybrid Optical-Electronic Edge Encoding for Privacy-Preserving Cloud Training"
  63rd ACM/IEEE Design Automation Conference (**DAC 2026**)

- He Zhu*, Xiaotong Huang*, Zihan Liu, Weikai Lin, Xiaohong Liu, Zhezhi He, Jingwen Leng, Minyi Guo, Yu Feng
  "SeeLe: A Unified Acceleration Framework for Real-Time Gaussian Splatting on Mobile Devices"
  IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2026**)
  CVPR 2026 Highlight

- Weikai Lin, Sushant Kondguli, Carll Marshall, Yuhao Zhu
  "PowerGS: Display–Rendering Power Co-Optimization for Neural Rendering in Power-Constrained XR Systems"
  SIGGRAPH Asia 2025 Conference Papers (**SIGGRAPH Asia 2025**)

- Weikai Lin*, Tianrui Ma*, Adith Boloor, Yu Feng, Ruofan Xing, Xuan Zhang, Yuhao Zhu
  "SnapPix: Efficient-Coding–Inspired In-Sensor Compression for Edge Vision"
  62nd ACM/IEEE Design Automation Conference (**DAC 2025**)

- Weikai Lin*, Yu Feng*, Yuhao Zhu
  "MetaSapiens: Real-Time Neural Rendering with Efficiency-Aware Pruning and Accelerated Foveated Rendering"
  30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS 2025**)
  ASPLOS 2025 Best Paper Award

- Yu Feng*, Zheng Liu*, Weikai Lin*, Zihan Liu, Jingwen Leng, Minyi Guo, Zhezhi He,

Jieru Zhao, Yuhao Zhu
"StreamGrid: Streaming Point Cloud Analytics via Compulsory Splitting and Deterministic Termination"
30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS 2025**)

- Yu Feng, Weikai Lin, Yuge Cheng, Zihan Liu, Jingwen Leng, Minyi Guo, Chen Chen, Shixuan Sun, Yuhao Zhu
"Lumina: Real-Time Mobile Neural Rendering by Exploiting Computational Redundancy"
52nd IEEE/ACM International Symposium on Computer Architecture (**ISCA 2025**)

- Adith Boloor, Weikai Lin, Tianrui Ma, Yu Feng, Yuhao Zhu, Xuan Zhang
"Private-Eye: In-Sensor Privacy Preservation Through Optical Feature Separation"
IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV 2025**)

- Nan Wu*, Weikai Lin*, Ruizhi Cheng*, Bo Chen, Yuhao Zhu, Klara Nahrstedt, Bo Han
"Advancing Immersive Content Delivery with Dynamic 3D Gaussian Splatting"
26th International Workshop on Mobile Computing Systems and Applications (**HotMobile 2025**)

- Yu Feng*, Weikai Lin*, Zihan Liu, Jingwen Leng, Minyi Guo, Han Zhao, Xiaofeng Hou, Jieru Zhao, Yuhao Zhu
"Potamoi: Accelerating Neural Rendering via a Unified Streaming Architecture"
ACM Transactions on Architecture and Code Optimization (**TACO 2024**)

- Wenfei Hu, Weikai Lin, Hongyu Fang, Yi Wang, Dingsheng Luo
"OW3Det: Toward Open-World 3D Object Detection for Autonomous Driving"
IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS 2024**)

- Wenfei Hu, Weikai Lin, Hongyu Fang, Yi Wang, Dingsheng Luo
"Learning Clear Class Separation for Open-Set 3D Detector in Autonomous Vehicle Via Selective Forgetting"
32nd IEEE International Conference on Robot and Human Interactive Communication (**RO-MAN 2023**)

- Weiming Qu, Tianlin Liu, Weikai Lin, Dingsheng Luo
"A Review of Robot Learning"
Acta Scientiarum Naturalium Universitatis Pekinensis, 59(6):1069–1086, 2023. DOI: 10.13209/j.0479-8023.2023.086

- Shuai Fang, Yaoyao Wei, Weikai Lin, Jianan Zhang, Tianlin Liu, Dingsheng Luo
"Approaching Sound Object with Sensorimotor Coordination when Sensors Partially Damaged"
IEEE International Conference on Development and Learning (**ICDL 2021**)

- Yaoyao Wei, Shuai Fang, Weikai Lin, Jianan Zhang, Dingsheng Luo
"Acquiring Robot Navigation Skill with Knowledge Learned from Demonstration"
IEEE International Conference on Development and Learning (**ICDL 2021**)

PREPRINTS AND WORKSHOPS

- Tianyu Hu, Weikai Lin, Weizhi Zhang, Jing Ma, Song Wang
"MemRouter: Memory-as-Embedding Routing for Long-Term Conversational Agents"
arXiv preprint arXiv:2605.00356, 2026

- Bo Yan, Weikai Lin, Yada Zhu, Song Wang
"SafeDream: Safety World Model for Proactive Early Jailbreak Detection"
arXiv preprint arXiv:2604.16824, 2026

- Weikai Lin*, Tianrui Ma*, Adith Boloor, Yu Feng, Ruofan Xing, Xuan Zhang, Yuhao Zhu
"SnapPix: Efficient-Coding–Inspired In-Sensor Compression for Edge Vision"

CVPR 2026 On-Sensor Vision (OSV) Workshop [link]

- Ruofan Xing, Arman Akbari, Weikai Lin, Adith Boloor, Alexander Montes McNeil, Michael Moebius, Yongmin Liu, Yuhao Zhu, Xuan Zhang
  "HoloCode: Hybrid Optical-Electronic Edge Encoding for Privacy-Preserving Cloud Training"
  CVPR 2026 On-Sensor Vision (OSV) Workshop [link]

- Han Yan, Yifei Zou, Weikai Lin, Xuan Zhang, Yuhao Zhu
  "Do Energy-Harvesting Pixels Reduce Carbon Cost of Visual Computing Systems?"
  HotEthics 2026 [link]

- Weikai Lin, Haoxiang Li, Yuhao Zhu
  "ControlHair: Physically-based Video Diffusion for Controllable Dynamic Hair Rendering"
  arXiv preprint arXiv:2509.21541, 2025

- Weikai Lin, Yu Feng, Yuhao Zhu
  "RTGS: Enabling Real-Time Gaussian Splatting on Mobile Devices Using Efficiency-Guided Pruning and Foveated Rendering"
  arXiv preprint arXiv:2407.00435, 2024 (preprint and shortened version of MetaSapiens)

PATENTS AND PROJECTS

- Dingsheng Luo, Xihong Wu, Yifan Yuan, Wenfei Hu, Weiming Qu, Yudi Zou, Jiawen Wang, Hongyu Fang, Weikai Lin
  Control Method and System for Improving Operation Precision of Robot Arm
  CN202310068025.4, China; published as CN116079730A, 2023 (**patent**)

- Dingsheng Luo, Xihong Wu, Shuai Fang, Jianan Zhang, Weikai Lin, Tianlin Liu
  Active Auditory Positioning Method for Map-Free Navigation
  CN202210079214.7, China; published as CN114563011A, 2022 (**patent**)

- Dingsheng Luo, Xihong Wu, Jianan Zhang, Shuai Fang, Tianlin Liu, Weikai Lin, Hongyu Fang
  Robot Map-Free Navigation Method Based on Time Sequence Information Modeling
  CN202110018866.5, China; published as CN112857370A, 2021 (**patent**)

- "Exploiting Human Color Discrimination for Memory- and Energy-Efficient Image Encoding in Virtual Reality: An FPGA Demo"
  Open-sourced project with optimized CPU/GPU/FPGA implementations. [code]

TEACHING EXPERIENCE

Teaching Assistant, University of Rochester

- Fall 2025, CSC 259/459 *Computer Imaging and Graphics*
- Spring 2025, CSC 252/452 *Computer Organization*
- Fall 2024, CSC 257/457 *Computer Networks*

EXTERNAL SERVICE

Reviewer

- ISMAR: 2026
- SIGGRAPH: 2025, 2026
- SIGGRAPH Asia: 2025
- WACV: 2026
- TCAD: 2025
- JCST: 2026